

Single Cycle RISC-V CPU Verification

The single-cycle and multi-cycle CPUs ran similar assembly programs to test their operations. The assembly programs tested each instruction of the RISC-V instruction set. The multi-cycle assembly program used memory address 0x2000 as the beginning of the RAM, while the single-cycle assembly program used memory address 0x0000 as the beginning of the RAM. For example, the multi-cycle assembly program used:

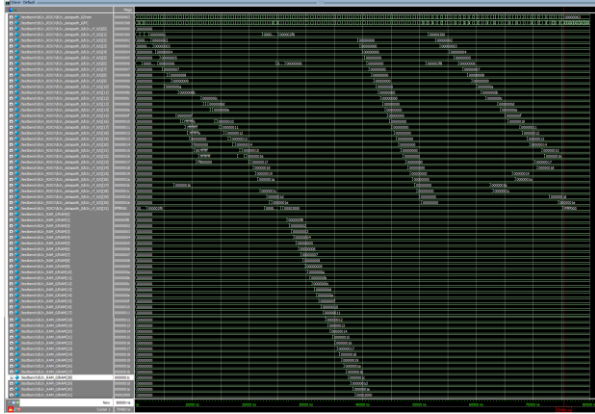
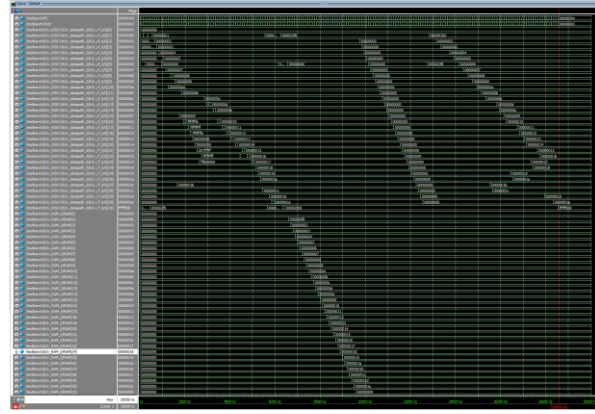
```
lb x2, 8(x31)
```

to load the first byte of RAM at address 0x2008, while the single-cycle assembly program used:

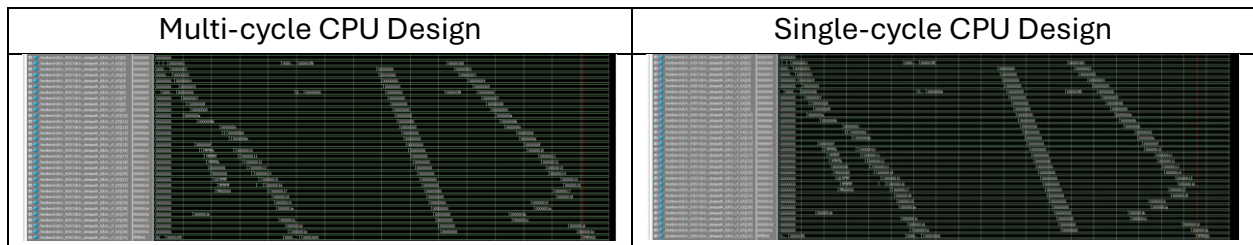
```
lb x2, 8
```

to load the same byte of RAM. The multi-cycle CPU design uses an address decoder to differentiate between RAM and ROM read/write operations. The single-cycle CPU design does not differentiate between RAM and ROM operations through the address.

The table below shows the wave windows produced by ModelSim after running each testbench for the CPU designs. Each wave window includes a red cursor (vertical line) indicating the end of the program. Each cursor has an associated timestamp, shown at the bottom. The cursors indicate that the multi-cycle CPU design finished executing the program in about 74.5 microseconds, while the single-cycle CPU design finished executing its program in about 18.6 microseconds. Therefore, the single-cycle CPU design is roughly four times faster than the multi-cycle CPU design.

Multi-cycle CPU Design	Single-cycle CPU Design
	
Multi-cycle Execution time: ~74.5 μs	Single-cycle Execution time: ~18.6 μs

The screenshots below show the register values of the CPU designs. The first column of each wave window shows the register number, Q[0] – Q[31], where the index inside the square brackets correspond to the register number. For example, Q[0] corresponds to register x0. Both CPU designs have the same register values at the end of the program, shown in the second column of each wave window.



The screenshots below show the RAM values of the CPU designs. The first column in each wave window shows the RAM location, RAM[0] – RAM[31]. Both CPU designs have the same RAM values at the end of the program, shown in the second column of each wave window.

